



Lighthief Cyprus Ltd

Renewable Energy & BESS Solutions

Battery Energy Storage System — Turnkey Proposal

Subarrow Investments Ltd — 2.64 MWp Solar Park, Agios Theodoros, Larnaca

Prepared for: **Antonis Maltezos — Subarrow Investments Ltd** February 2026

Ref: LCY-IND-MALTEZOS-2026-02 Commercial — Confidential

2.5

MW

BESS POWER
RATING

10

MWh

ENERGY
STORAGE
CAPACITY

€1,447,800

TURNKEY INVESTMENT
(EX VAT)

~25%

YEAR 1 ROI

1. Executive Summary

This proposal presents a **turnkey battery energy storage system (BESS)** designed for your 2.64 MWp solar park in Agios Theodoros, Larnaca — planned to energize in **December 2026**.

Every solar park in Cyprus today faces **curtailment** — periods when the grid operator (DSO) forces your park to reduce or stop production because the grid has too much solar power. When your park is curtailed, your solar panels are producing electricity, but you are **not allowed to sell it**. That energy is destroyed.

A BESS solves this by **capturing curtailed energy that would otherwise be wasted** and storing it in batteries. The stored energy is then discharged and sold during the **high-price evening window** when electricity is most valuable.

Bottom line: Your park will produce ~4,752 MWh per year. With a projected 55% curtailment rate, roughly **€368,000/year** of your energy will be wasted — unless you install a BESS. This system recovers that lost revenue and turns it into **€360,000 annual net income**, paying for itself in just 4 years.

2. The Problem: Solar Curtailment in Cyprus

2.1 What Is Curtailment?

Curtailment occurs when the **Cyprus DSO (Distribution System Operator)** orders your solar park to reduce or stop feeding electricity into the grid. This typically happens during peak sunshine hours (10:00–14:00) when too many solar parks are generating at the same time and the grid cannot absorb all the power.

When you are curtailed, **your park is producing electricity but you earn zero revenue from it**. The energy is simply lost.

2.2 Why Curtailment Is Increasing

Cyprus has been rapidly adding solar capacity. The TSOC Day-Ahead Market data shows that **42.5% of all trading days** (57 out of 134 days since the market launched) have had periods where the electricity price dropped to **€0/MWh** during solar hours. This is a clear signal of growing oversupply during daytime — exactly when your park produces.

With more solar parks coming online every quarter, curtailment rates are widely expected to **increase from current levels**. Parks energizing in late 2026 will face higher curtailment than those that started years ago.

2.3 What This Means for Your Park

Curtailment Rate	Energy Wasted (MWh/yr)	Revenue Lost Per Year
30%	1,426	€200,800
40%	1,901	€267,800
50%	2,376	€334,700
55% (projected baseline)	2,614	€368,300
60%	2,851	€401,700

At 55% curtailment, your park loses **€368,300 every year** — energy your panels produce but you cannot sell. Over 15 years, that is **€5.3 million** in permanently destroyed revenue.

3. The Solution: Battery Energy Storage (BESS)

3.1 How a BESS Makes Money

A BESS is a large-scale battery system installed next to your solar park. It works in a simple cycle every day:



3.2 Why the Economics Are So Attractive

Factor	Value
Cost to charge the battery	€0/MWh — this is curtailed energy you'd lose anyway
Price when you discharge	€182.99/MWh — verified peak evening average
Net revenue per MWh stored	€157.96/MWh (after 86.32% round-trip efficiency)

Charge at €0, discharge at €183. Because you are storing your own curtailed energy (not buying from the grid), the BESS effectively creates revenue from nothing. Every kWh stored is a kWh that would have been destroyed.

Important regulatory note: As of February 2026, BESS in Cyprus cannot buy electricity from the DAM grid — revenue is limited to curtailment recovery only. When future legislation enables grid arbitrage, your system will generate **additional revenue** not included in this financial model.

4. Your Solar Park

Parameter	Value
SPV	Subarrow Investments Ltd
CERA License	E004576/2024
Installed Capacity	2.64 MWp
Location	Agios Theodoros, Larnaca District
Planned Energization	December 2026
Technology	Bifacial TopCon PV modules, fixed tilt south
Annual Yield	1,800 kWh/kWp
Annual Production	4,752 MWh
Projected Curtailment	55% (baseline estimate for 2027 operation)
Projected Curtailed Energy	2,614 MWh/year
Site Visit	Completed by Lighthief engineering team

5. Proposed System — 2.5 MW / 10 MWh

Turnkey System Price

€1,447,800 ex VAT

€144,780 per MWh of storage capacity

Power Rating	2.5 MW
Energy Capacity	10.03 MWh (2 × 5.015 MWh containers)
Discharge Duration	4 hours at rated power
Power Match to Park	94.7% (2.5 MW / 2.64 MW) — near-perfect

5.1 What's Included in the Turnkey Price

Item	Status
2 × Linyang LFP battery containers (5.015 MWh each)	✓ Included
MV Skid (PCS inverter + transformer)	✓ Included
CIF delivery — Shanghai to Limassol port	✓ Included
Import duties & customs clearance	✓ Included
Inland transport & crane to site	✓ Included
Civil works — concrete bases, drainage, fencing	✓ Included
LV/MV cabling & terminations	✓ Included
Internal surge protection (SPDs) & earthing	✓ Included
Protection engineering & relay coordination	✓ Included
Voltus Energy Management System (EMS)	✓ Included
SCADA — local site + global monitoring	✓ Included
Installation & commissioning (OEM crew on site)	✓ Included
ETEK engineer certification	✓ Included
EU Battery Passport compliance documentation	✓ Included
Construction insurance (CAR/EAR)	✓ Included
Compliance & safety documentation	✓ Included
Project management & coordination	✓ Included

5.2 Excluded (Client-Paid)

Item	Notes
EAC testing & connection fees	Paid directly by park owner to EAC
External Lightning Protection (LPS)	DEHN system — coordinated by Lighthief
Communication equipment to TSO/DSO	Paid directly by park owner
VAT (19%)	Applied on the EPC price

6. System Specifications

Parameter	Specification
Rated Power	2.5 MW
Rated Energy	10.03 MWh (2 × 5.015 MWh containers)
Discharge Duration	4 hours at rated power
Round-Trip Efficiency	86.32% AC-AC (full system incl. cabling losses)
Battery Cells	EVE Energy — LFP (Lithium Iron Phosphate)
PCS Inverter	Kehua BCS1000K / BCS1250K
System Integrator	Linyang Energy (Shanghai-listed, \$3.5B assets)
Cycle Life	7,000 cycles at 70% end-of-life capacity
Grid Code Compliance	EN 50549-2 (TÜV cert D 115067 0077)
Corrosion Rating	C5 (Cyprus coastal/inland environments)
Delivery Terms	CIF Limassol (Incoterms 2020)

7. Without BESS vs With BESS — The Core Comparison

This is the central question: what happens to your park's revenue over 15 years **with and without** a battery system? The model below uses a **projected 55% curtailment rate** as the baseline for a park energizing in late 2026.

7.1 Year 1 Revenue Comparison

✗ Without BESS

-€368,300

lost to curtailment every year

Your park produces 4,752 MWh

You sell only 2,138 MWh (45%)

2,614 MWh is wasted — gone forever

✓ With BESS

+€360,000

net income from recovered energy

BESS captures 2,483 MWh of curtailed energy

Discharges 2,143 MWh at peak prices

Gross revenue: €392,200 (minus €32,200 O&M)

7.2 Revenue Mechanics

Step	Value	Explanation
1. Curtailed Energy	2,614 MWh/yr	55% of 4,752 MWh annual production
2. BESS Capture Rate	95%	2.5 MW covers 94.7% of park power; 10 MWh exceeds daily need
3. Energy Captured	2,483 MWh/yr	Stored in the battery during curtailment
4. Round-Trip Efficiency	86.32%	Energy lost to heat during charge/discharge cycle
5. Energy Discharged	2,143 MWh/yr	Actually sold to the market
6. Peak Evening Price	€182.99/MWh	Verified average, 17:00–21:00, 134 TSOC days
7. Gross Revenue	€392,200/yr	2,143 MWh × €182.99
8. Annual O&M	(€32,200/yr)	Maintenance, monitoring, insurance
9. Net BESS Income	€360,000/yr	Pure additional income to your park

7.3 15-Year Comparison

✗ Without BESS — 15 Years €0 from curtailed energy ~€5.3M in solar energy destroyed Your panels produce it. You cannot sell it. Permanent, irrecoverable loss.	✓ With BESS — 15 Years +€2,863,000 pure profit from recovered energy Invest €1,447,800 → Earn €4,311,000 net Investment recovered in ~4 years Then 11+ years of pure profit
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8. Investment Return Summary

All figures based on **55% projected curtailment**. See Section 9 for returns at different curtailment rates.

Metric	Value
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BESS Investment	€1,447,800
Year 1 Net Income	€360,000
Year 1 ROI	24.9%
Simple Payback	4.0 years
15-Year Net Income	€4,311,000
15-Year Profit	€2,863,000
15-Year Cash-on-Cash	298%

8.1 Year-by-Year Projection (55% Curtailment Baseline)

Battery capacity degrades ~2.5%/year. PV output degrades 0.5%/year. O&M held constant at €32,200/year. Combined effect: ~3% annual decline in discharged energy.

Year	PV (MWh)	Curtailed	Discharged	Revenue	O&M	Net Income	Cumulative
1	4,752	2,614	2,143	€392,200	€32,200	€360,000	€360,000
2	4,728	2,600	2,079	€380,400	€32,200	€348,200	€708,200
3	4,705	2,588	2,017	€369,100	€32,200	€336,900	€1,045,100
4	4,681	2,575	1,956	€357,900	€32,200	€325,700	€1,370,800
5	4,658	2,562	1,897	€347,100	€32,200	€314,900	€1,685,700
6	4,635	2,549	1,840	€336,700	€32,200	€304,500	€1,990,200
7	4,612	2,537	1,785	€326,600	€32,200	€294,400	€2,284,600
8	4,589	2,524	1,731	€316,800	€32,200	€284,600	€2,569,200
9	4,566	2,511	1,679	€307,200	€32,200	€275,000	€2,844,200
10	4,543	2,499	1,629	€298,000	€32,200	€265,800	€3,110,000
11	4,521	2,487	1,580	€289,100	€32,200	€256,900	€3,366,900
12	4,498	2,474	1,533	€280,500	€32,200	€248,300	€3,615,200
13	4,476	2,462	1,487	€272,100	€32,200	€239,900	€3,855,100
14	4,453	2,449	1,442	€263,900	€32,200	€231,700	€4,086,800
15	4,431	2,437	1,399	€255,900	€32,200	€223,700	€4,310,500

9. Curtailment Scenario Analysis

Since your park will energize in December 2026, the actual curtailment rate is not yet known. The table below shows how your BESS investment performs across a range of curtailment scenarios — from a conservative 30% to a high 65%.

Curtailment Rate	Curtailed (MWh/yr)	Net Income (Y1)	ROI	Payback
30% (conservative)	1,426	€181,700	12.6%	8.0 years
35%	1,663	€217,400	15.0%	6.7 years
40%	1,901	€253,100	17.5%	5.7 years
45%	2,138	€288,600	19.9%	5.0 years
50%	2,376	€324,300	22.4%	4.5 years
55% (projected baseline)	2,614	€360,000	24.9%	4.0 years
60%	2,851	€395,600	27.3%	3.7 years
65%	3,089	€431,200	29.8%	3.4 years

The investment is profitable across all realistic scenarios. Even at a conservative 30% curtailment, the BESS delivers a 12.6% annual return — far exceeding bank deposit or bond yields. At the projected 55% baseline, the system pays for itself in just 4 years with a 24.9% ROI.

Industry trend: Curtailment rates in Cyprus have been trending upward. As more solar capacity connects to the grid, the same amount of grid capacity must absorb more energy — leading to more frequent and deeper curtailment events. Your BESS becomes **more valuable** over time as curtailment increases.

10. Verified Market Data

Source: 134 TSOC DAM Excel files | 6,432 half-hourly records | 1 Oct 2025 – 11 Feb 2026
Download: tsoc.org.cy — [DAM Reports](#)

Metric	Verified Value
Average MCP (all periods)	€158.19/MWh
Solar Hours Average (06:00–17:00)	€140.88/MWh
Midday Average (10:00–14:00)	€101.13/MWh
Peak Evening Average (17:00–21:00)	€182.99/MWh
Peak–Midday Spread	€81.86/MWh
Days with positive spread	134 / 134 (100%)
Days hitting zero price	57 / 134 (42.5%)

Month	Days	Avg MCP	Midday Avg	Peak Avg	Spread
Oct 2025	31	€147	€79	€178	€99
Nov 2025	30	€155	€82	€185	€103
Dec 2025	31	€159	€111	€179	€67
Jan 2026	31	€169	€138	€188	€50
Feb 2026	11	€164	€85	€192	€107

The **peak evening price has ranged from €178 to €192/MWh** and has been higher than midday in 100% of trading days. This price spread is the foundation of BESS profitability — and it is expected to widen as solar penetration increases.

11. Future Revenue Upside — Grid Arbitrage

When future legislation enables BESS to buy electricity from the grid (expected within 1–3 years), your system gains **additional income** on days when curtailment is low or absent:

Scenario	Additional Annual Revenue
100 arbitrage days/year × 10 MWh	€56,630
200 arbitrage days/year × 10 MWh	€113,260

Future grid arbitrage could add **€57K–113K/year** on top of curtailment recovery. This is **not included** in the financial model — it represents pure upside that would significantly accelerate your payback.

12. Warranties & Performance Guarantees

Guarantee	Term
Base Warranty (defects, workmanship)	5 years from commissioning
Extended Warranty — BESS (Years 6–10)	€913.92/MWh/year
Extended Warranty — BESS (Years 11–15)	€1,157.62/MWh/year
Extended Warranty — PCS+MVS (Years 6–10)	€747.76/MWh/year
Extended Warranty — PCS+MVS (Years 11–15)	€926.10/MWh/year
Combined Extended (Yr 6–10)	€1,661.68/MWh/year
Combined Extended (Yr 11–15)	€2,083.72/MWh/year
Battery Health (SOH) — Year 5	≥ 85%
Battery Health (SOH) — Year 10	≥ 79.58%
Battery Health (SOH) — Year 15	≥ 70%
System Round-Trip Efficiency	≥ 86.32% AC-AC
PCS Efficiency	≥ 98% at rated power
System Availability (with LTSA)	≥ 97%

What this means in plain terms: The first 5 years are fully covered under warranty at no extra cost. After that, you can purchase extended warranty coverage which guarantees the battery maintains its performance. If the battery health drops below the guaranteed levels, the manufacturer is contractually obligated to remedy it.

13. Long-Term Service Agreement (LTSA)

An LTSA is an ongoing maintenance and monitoring contract that keeps your BESS performing optimally. Think of it as a comprehensive service plan — similar to a car maintenance contract, but for your battery system.

Service	Rate (€/MWh/yr)	Annual (10 MWh)
BESS Preventive & Corrective Maintenance	€1,157.62	€11,576
PCS + MVS Maintenance	€1,311.97	€13,120
Total LTSA	€2,469.59	€24,696

Service Level	Detail
Provider	Lighthief Cyprus Ltd
Response Time	4 hours maximum
Monitoring	24/7/365 remote monitoring via SCADA
Availability Guarantee	97% annual uptime
Spare Parts	Held locally in 950 m² warehouse, Limassol

14. Annual Operating Costs

These are the ongoing costs already factored into the financial model above. They are deducted from gross revenue to calculate your net income.

Cost Item	Annual Cost
BESS Preventive & Corrective Maintenance	€11,576
PCS + MVS Maintenance	€13,120
Insurance (operational, 0.75% of equipment value)	€7,507
Total Annual O&M	€32,203

15. Payment Terms & Project Timeline

15.1 Payment Schedule

Milestone	%	Amount	Timing
Advance Payment	25%	€361,950	At contract signing
Ready to Ship	50%	€723,900	Equipment ready, factory inspection passed
On-Site Delivery	20%	€289,560	Equipment arrives at your site
SAT Complete	5%	€72,390	After commissioning & testing pass
Total	100%	€1,447,800	

15.2 Delivery Timeline — Aligned with Park Energization

Your solar park is planned to energize in **December 2026**. To have the BESS operational at the same time, we recommend placing the order by **April 2026**:

Phase	Duration	Target Dates
Contract signing & advance payment	—	April 2026
Factory production	~13 weeks	April – June 2026
Sea transport (Shanghai → Limassol)	~7 weeks	July – August 2026
Customs, inland transport & site delivery	~2 weeks	September 2026
Installation & commissioning	4–6 weeks	September – October 2026
Testing, EAC approval & buffer	~4 weeks	November 2026
BESS Operational	~8 months	December 2026

Perfect alignment: By ordering in April, your BESS will be installed, commissioned, and ready to operate when your solar park energizes in December 2026. From day one, every curtailed MWh is captured — not a single day of wasted energy.

16. Why Lighthief Cyprus

Capability	Detail
Established	2015 — over a decade in renewables
Global Presence	11 countries across Europe and Asia
Team	150+ engineers and specialists
R&D Center	23-hectare facility in Poland (2×8 MW PV, BESS lab)
Cyprus BESS Pipeline	249 MW / 882 MWh across 51 parks — currently deploying
OEM Partner	Linyang Energy (Tier-1, Shanghai-listed, 7,000-cycle warranty)
EMS/SCADA	Voltus Energy Management Systems
Local Presence	Limassol office, 950 m² spare parts warehouse
Site Assessment	Agios Theodoros site visit completed

17. Important Notes & Disclaimers

1. Market prices are historical averages from the first 134 days of the Cyprus competitive electricity market. Future prices may vary.
2. The 55% curtailment rate is a **projected estimate** for a park energizing in late 2026. Actual curtailment will depend on DSO dispatch-down orders and grid conditions in the Larnaca region. See Section 9 for returns at different curtailment levels.
3. PV yield (1,800 kWh/kWp) is based on the park design specification for bifacial TopCon modules at fixed tilt south in Larnaca. Actual yield may vary.
4. The capture rate (95%) is an estimate based on the 2.5 MW / 2.64 MW power ratio and the 10 MWh energy capacity relative to projected daily curtailment. Actual capture depends on daily curtailment distribution.
5. BESS regulatory status is current as of February 2026. Grid arbitrage requires future legislation.
6. All figures exclude VAT (19%), grid connection costs, and land-related expenses.
7. Battery degradation (~2.5%/year) is conservative. Actual degradation depends on cycling, temperature, and maintenance.
8. This proposal is valid for 30 days from the date of issue.

Contact & Next Steps

Turn €368,000/year of wasted energy into €360,000/year of income — from Day 1.

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Recommended Next Steps:

1. Review this proposal and the curtailment scenarios
2. Schedule a call to discuss any questions
3. Contract signing and advance payment (recommended by April 2026)
4. Production begins — BESS operational by December 2026, aligned with your park

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